CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The main source of heat and light for the Earth is the: 1

(a) ocean **(b)** Sun **(c)** volcano **(d)** Moon

2. Heat from the Sun reaches the Earth by: 1

(a) conduction **(b)** convection **(c)** evaporation **(d)** radiation

3. Woollen clothes keep us warm because they: 1

(a) trap air, a poor conductor **(b)** are shiny

(c) produce heat **(d)** conduct heat well

4. Smoke rises from a burning fire because: 1

(a) smoke is heavy **(b)** hot air rises **(c)** wood is light **(d)** air is cold

5. Water gets heated in a vessel from the bottom by: 1

(a) radiation only **(b)** convection currents

(c) conduction only **(d)** sublimation

6. The continuous movement of water between land, sea and atmosphere is the: 1

(a) food chain **(b)** rock cycle **(c)** water cycle **(d)** carbon cycle

7. The seeping of surface water through soil and rocks is: 1

(a) condensation **(b)** infiltration **(c)** precipitation **(d)** evaporation

8. The SI unit of temperature used in everyday life in India is: 1

(a) metre **(b)** second **(c)** degree Celsius **(d)** kilogram

9. Ice kept in a sawdust box melts slowly because sawdust is: 1

(a) wet **(b)** a poor conductor of heat

(c) hot **(d)** a good conductor

10. Which item will absorb the most heat from a campfire? 1

(a) A mirror **(b)** A black cloth **(c)** A white cloth **(d)** A shiny foil

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What is conduction? Give an example. 2

12. What is convection? Give an example. 2

13. What is radiation? How does heat reach us from the Sun? 2

14. Why do we wear woollen clothes in winter? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show conduction through a metal rod. 3

16. Describe an activity to show convection in water. 3

17. Explain sea breeze and land breeze. 3

OR

Explain the water cycle with the help of a diagram (describe). 3

18. Differentiate between conduction, convection and radiation. 3

OR

Why are cooking vessels made of metals with plastic handles? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Name the three modes of heat transfer. 5
(ii) Explain each with an example.
(iii) Which needs no medium?

OR

Case study: Pema and Palden sit around a campfire; smoke rises; they feel warm, and a metal rod in the fire gets hot at the handle. 5
(i) Modes at work.
(ii) Why does smoke rise?
(iii) Why is the handle hot?
(iv) How do they feel warmth?

20. (i) Describe the water cycle. 5
(ii) What role does the Sun play?
(iii) How is groundwater formed?

OR

Case study: At the beach, winds blow from the sea during the day and in the reverse direction at night. 5
(i) Names of winds.
(ii) Reason for day.
(iii) Reason for night.
(iv) Who benefits?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND